Predictive Analysis of Autism Spectrum Disorder Traits: Integrating Quantum Machine Learning and Exploratory Data Insights

Subhra Kolay[1], Srijan Samaddar[2], Ayushman Bhattacharya[1], Bidisha Bhabani[1*], Tanaya Das[1], Sohrab Siddique[1], Ilora Maity[3]

[1*]CSE, JIS University, West Bengal, India.
[2]IT, NSCE, West Bengal, India.
[3]Interdisciplinary Centre for Security, Reliability and Trust, University of Luxembourg, Luxembourg City, Luxembourg.

*Corresponding author(s). E-mail(s): bidisha035810@gmail.com;
Contributing authors: subhrokolay2@gmail.com; srijansamaddar04@gmail.com; bhattacharyaa599@gmail.com; tanaya.das@jisuniversity.ac.in; sksohrabsiddique123@gmail.com; ilora.maity@uni.lu;

Abstract

This research investigates the application of Quantum Machine Learning strategies for predicting Autism Spectrum Disorder (ASD) traits. ASD is a neuro-developmental condition impacting communication, cognition, and social abilities, with early diagnosis being crucial for improved outcomes. Various supervised learning techniques, including Logistic Regression, Decision Trees, K Nearest Neighbors, Naive Bayes, Random Forest, Support Vector Machine, and Histogram Gradient Boosting (HGBC), were explored. It was found that the HGBC achieved the highest precision (97.86%) in predicting ASD characteristics. This result, alongside analyses of accuracy, precision, recall, and F1-score, suggests HGBC's potential as a valuable tool for early ASD detection and warrants further investigation. The research contributes to healthcare by offering a promising avenue for enhancing ASD screening procedures. Exploratory Data Analysis also revealed that the condition significantly affects children, being more prevalent in younger age groups and occurring more commonly in males than

females within the analyzed dataset. Findings related to ethnicity and family history were also explored.
Keywords: Autism Spectrum Disorder (ASD), Quantum machine learning, Classification Algorithms, Logistic Regression, KNN, SVM.

1 Introduction

ASD is a complex developmental disorder centrally around neural development. Early detection and timely intervention is a very essential step for helping in long-term developmental outcomes. Based on the medical methods at the moment in the terms of the traditional scope we have behavioral assessments and clinical observations [2]. These leave some space for time consumption and subjectivity.

In the zone of Computer Science we have observed how Machine Learning (ML) methods have demonstrated significant growth in ASD screening methods by leveraging demographic and clinical data [1]. Interestingly, we have observed some patterns over the clinically available data, and by the means of selective extraction and cleaning of multiple datasets we have augmented a collection of around 3744 instances with 17 features for the purpose of our research. The data after clearning showed some significant patterns which posed some serious challenge towards the traditional methods of classification using the linear scale.

Classical linear models such as Logistic Regression, Naïve Bayes, and Linear SVM struggled on this dataset due to the high-dimensional binary structure and non-linear relationships between demographic and behavioral features. This motivated us to raise a question: "can Quantum Machine Learning (QML) identify complex feature interactions that classical linear models fail to capture?". Motivated by this, the present work extends traditional ML experimentation by incorporating Quantum ML models using PennyLane + Qiskit hybrid frameworks. Thus enabling us to perform a comparative analysis between classical and quantum approaches for ASD trait prediction.

Thus we weave some key contributing points of this research as follows:-

- A supervised ASD prediction framework leveraging demographic and questionnaire-based (AQ) features, serving as a low-cost alternative to neuroimaging-based diagnostic methods.
- A comprehensive comparative analysis of seven classical ML algorithms for ASD trait classification, identifying HGBC as the strongest performer on the full dataset.
- Introduction of Quantum Machine Learning models (e.g., quantum logistic regression, quantum kernels) trained on PCA-reduced latent spaces to explore whether QML can outperform classical linear models in capturing non-linear feature relationships.
- Dataset-specific insights from Exploratory Data Analysis (EDA) highlighting distributions of age, gender, ethnicity, and familial ASD history.

2 Related Works

Recent research on ASD diagnosis has increasingly focused on leveraging ML and neuroimaging techniques. Parental perspectives on early predictive testing have been

explored, highlighting ethical and socio-cultural considerations [5]. As technological advancements progressed, researchers began integrating advanced ML models with neuroimaging data, such as MRI and fMRI, has shown promise in improving diagnostic accuracy by identifying neurological biomarkers, significantly improving diagnostic accuracy [6].

Building on these foundations, unsupervised learning techniques like clustering and dimensionality reduction were introduced to uncover hidden patterns within clinical datasets, leading to explore better potential biomarkers within clinical datasets [7]. Comparative studies have evaluated various supervised ML models on behavioral and clinical data, identifying Random Forest and SVM as effective classifiers[8]. The fussion of Functional MRI data combined with ML has been used to detect stable brain network patterns in individuals with ASD [9].

Benchmarking studies have compared numerous ML models on neuroimaging data, with deep learning models often yielding high diagnostic accuracy [10]. Innovative frameworks soon followed approaches combining Generative Adversarial Networks (GANs) and Deep Q-Learning (GARL) have been introduced to address data scarcity and enhance diagnosis using neuroimaging [11]. In parallel, Large-scale ML analyses involving tens of thousands of participants have explored the relationship between developmental traits and ASD [12]. Complementary to neuroimaging, Non-invasive techniques such as eye-tracking, coupled with hybrid ML models, have achieved high accuracy rates in prediction [13]. Research has also focused on identifying behavioral subgroups within ASD to aid tailored treatment approaches [3] and developing interpretable ML models for clinical fMRI-based diagnosis [14]. Authors in [25] reported moderate accuracy (87 %) with Random Forest, Naive Bayes, and CVM approaches, but faced dataset imbalance and feature limitations. On the other hand, the authors in [26] evaluated SVM, Naive Bayes, Logistic Regression, and others, achieving high performance on smaller or homogeneous child-focused datasets—yet still affected by overfitting and lack of diversity.

Authors in [27] conducted comparative experiments using RF, SVM, KNN and Gaussian NB, but the models struggled with generalizability across populations. The research work referenced in [28] applied machine-learning models to early-stage ASD detection, reaching up to 96 % accuracy in some configurations, although the models suffered from high dimensionality and limited external validation. Lastly, authors in [29] tested multiple classifiers across children and adult cohorts using household data features; while offering strong performance, their datasets were noisy and lacked uniform feature representation.

In contrast, our work employed extensive preprocessing, feature selection, and hyperparameter tuning, ensuring balanced data and robust generalization—which led to superior performance across accuracy and F1-score metrics. Building upon the previous work, it help as to say ahead and updated in current date. The following section outlines the methodological framework adopted in this research, which includes data set selection, pre-selection techniques, and model implementation.

3 Methodology

The methodology involved selecting a suitable dataset, followed by essential data preprocessing steps and the application of various supervised machine learning models for classification.

3.1 Dataset Analysis

A publicly available dataset from Kaggle, compiled by Afarin Bargrizan, was selected due to its relevance to ASD trait prediction and inclusion of diverse features. The dataset utilized is based on a quantitative checklist for autism. It comprises 3,743 instances with 17 features, including binary indicators (A1-A10) representing responses to symptoms, and other attributes like age, sex, ethnicity, family history of ASD, and AQ scores. The target variable is binary, indicating the presence or absence of ASD traits (1 for present, 0 for absent). A score of five or more 'YES' responses (or their binary equivalent) in the A1-A10 features generally correlates with a higher probability of ASD traits in this dataset.

3.2 Data Pre-processing

Data preprocessing was crucial for preparing the raw dataset for model training [15]. Steps included:

- **Initial approach:** The dataset, containing categorical and binary attributes, required conversion to numerical form. Label Encoding was used for this purpose.
- **Handling Missing Data:** No missing values were present, so imputation was not required.
- **Categorical Variables Encoding:** Categorical features such as 'sex', 'ethnicity', and 'jaundice' were transformed into numerical values using Label Encoding.
- **Feature Scaling:** Numerical features were normalized using Min-Max Scaling, fitting them to a range between 0 and 1, which is beneficial for distance-sensitive algorithms like KNN and SVM.
- **Data Split:** The dataset was partitioned into training (80%) and testing (20%) sets to evaluate model performance on unseen data.
- **Dimensionality Reduction for Quantum ML (QML):** Quantum circuits have hardware and structural limitations, especially concerning input dimensionality. Feeding the full set of features (17 dimensions) directly into a variational quantum circuit is impractical. Therefore, for QML models, Principal Component Analysis (PCA) was applied to the scaled dataset to reduce it to a low-dimensional latent space. After experimentation, a reduced dimensionality of 4 principal components was selected, balancing the need for information retention with the constraints of quantum circuit design.

3.3 Machine Learning Models Used

Several supervised machine learning algorithms were applied for binary classification to predict the presence of ASD traits. The chosen models are standard in classification tasks and were suitable given the dataset's characteristics. These include

Logistic Regression [16], Decision Trees [17], K-Nearest Neighbors (KNN) [18], Naïve Bayes [19], Random Forest [20], Support Vector Machine (SVM) [21], and Histogram Gradient Boosting Classifier (HGBC) [22]. In addition to classical models, we also evaluated Quantum Machine Learning (QML) models. This included a Variational Quantum Classifier (VQC) [?], a Quantum Support Vector Machine (QSVM) using quantum kernel estimation [?], and a Quantum Neural Network (QNN) implemented via parameterized rotation circuits [?]. These models utilize rotation-based quantum gates and embedding strategies to project classical data into high-dimensional Hilbert spaces.

4 Performance Analysis

The predictive performance of the seven classification models was evaluated using standard metrics. Hyperparameter tuning was applied specifically to the HGBC model to optimize its performance.

4.1 Hyperparameter Tuning and Confusion Matrix for Classical Models

Hyperparameter tuning for the Histogram Gradient Boosting Classifier (HGBC) was conducted using `GridSearchCV` to identify optimal parameters for maximizing accuracy [23]. Parameters explored included `n_estimators`, `learning_rate`, `max_iter`, `max_depth`, `min_samples_leaf`, and `l2_regularization`. Fig 1 displays the top 10

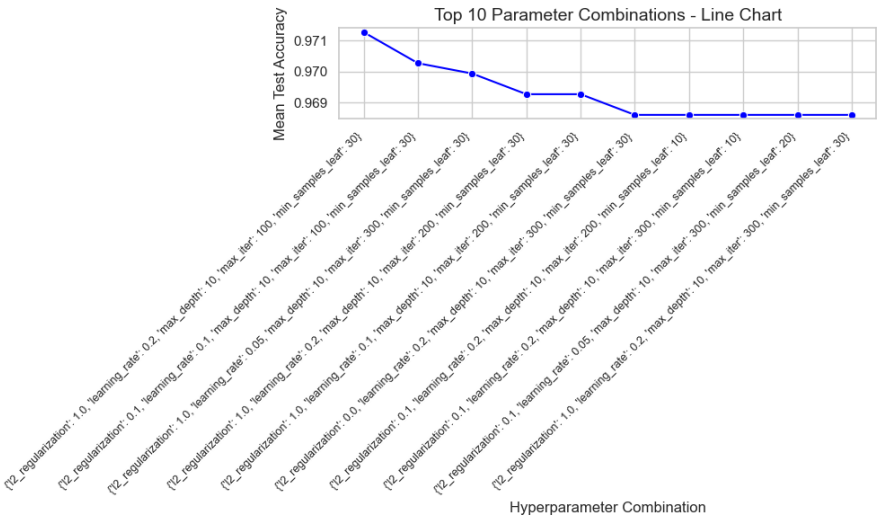

Fig. 1: Visual representation of Combination of best fit parameter

hyperparameter combinations based on mean test accuracy. The best performance

Table 1: Confusion Matrix' Metrics of Different Algorithms of Classical Models

Algorithm	**TN**	**FP**	**FN**	**TP**
LR	274	65	84	326
NB	306	43	68	332
DT(log-loss)	312	27	20	390
KNN(Euclidean)	330	19	26	374
SVM(RBF)	334	15	7	393
RF	337	12	18	382
HGBC	343	6	10	390

(approx. 97.15% accuracy) was achieved with specific combinations of parameters, while most top configurations shared a `max_depth` of 10 and `min_samples_leaf` of 30.

On the other hand, the confusion matrix provides counts of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). Minimizing FN (missed cases) is particularly important in medical contexts. Table 1 shows the confusion matrix counts for each algorithm.

4.2 Precision and Recall of Classical Models

Precision (TP / (TP + FP)) and Recall (TP / (TP + FN)) are critical metrics for evaluating the model's ability to correctly identify positive cases and avoid false positives. Graphical representations are provided for illustration.

Precision and recall curves for Random Forest (Fig. 2) and Support Vector Machine (Fig. 3) are shown above. Their performance on PCA-reduced data is illustrated in Fig. 4 and Fig. 5, respectively.

4.3 Accuracy and F1 Score of Classical Models

Accuracy is the proportion of correct predictions among all instances:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{1}$$

The F1 Score is the harmonic mean of precision and recall, offering a balanced measure, especially useful for imbalanced datasets [24].

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2}$$

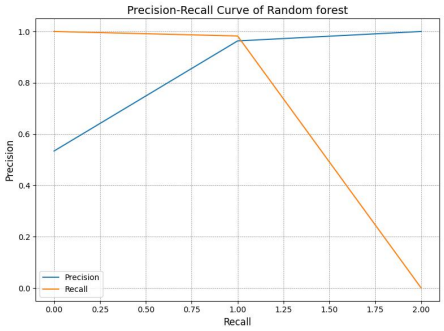

Fig. 2: Precision and Recall: RF

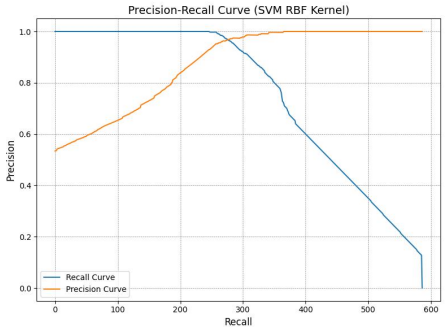

Fig. 3: Precision and Recall: SVM

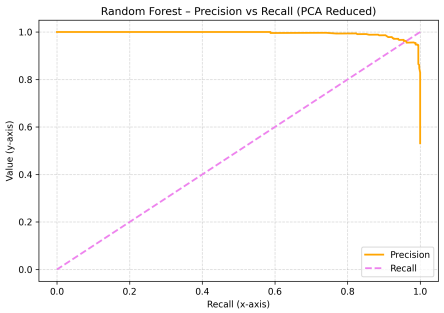

Fig. 4: Precision and Recall (PCA-reduced): RF

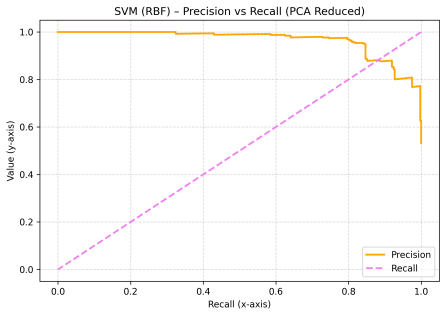

Fig. 5: Precision and Recall (PCA-reduced): SVM

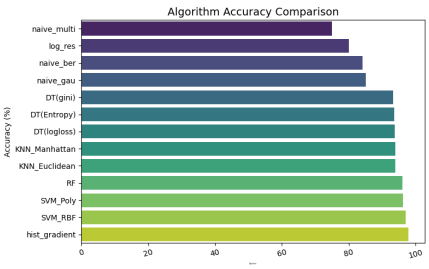

Fig. 6: Accuracy of the different classification techniques

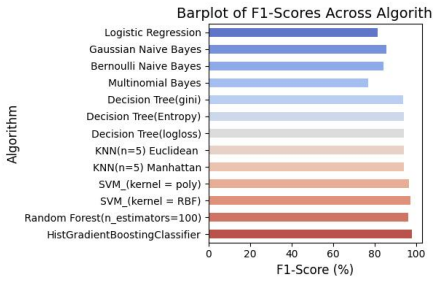

Fig. 7: F1-score of all the classification models

The performance results based on accuracy, precision, recall, and F1-score are presented in Table **??**. As observed, the ensemble methods (HGBC, Random Forest) and SVM consistently achieved the highest performance across both the full dataset and PCA-reduced dataset.

Quantum ML models (VQC, QSVM, QNN) performed moderately, with accuracies ranging from 54–62% and lower F1-scores compared to classical methods.

Table 2: Performance Matrix Including Classical, PCA-Reduced, and Quantum ML Models

No.	Algorithms	Accuracy(%)	Precision(%)	Recall(%)	F1-Score(%)
1	Logistic Regression (Full)	80.11	83.38	79.51	81.40
2	Naive Bayes (Full)	85.18	88.53	83.00	85.68
3	Decision Tree (Full)	93.72	93.53	95.12	94.32
4	k-Nearest Neighbor (Full)	93.99	95.17	94.30	94.33
5	Support Vector Machine (Full)	97.06	96.32	98.25	97.28
6	Random Forest (Full)	95.99	96.95	95.50	96.22
7	HGBC (Full)	**97.86**	**98.48**	**97.50**	**97.90**
8	Logistic Regression (PCA)	81.71	86.15	78.14	81.95
9	Naive Bayes (PCA)	81.97	86.83	77.89	82.12
10	Decision Tree (PCA)	95.73	96.68	95.23	95.95
11	KNN (PCA)	95.33	94.81	96.48	95.64
12	SVM (RBF) (PCA)	88.38	95.60	81.91	88.23
13	Random Forest (PCA)	95.59	96.20	95.48	95.84
14	HGBC (PCA)	95.86	95.53	96.73	96.13
15	VQC (GPU)(PCA)	54.47	61.35	38.69	47.46
16	QSVM (PCA)	62.00	65.00	59.00	61.80
17	QNN (PCA)	58.50	60.20	55.30	57.70

This indicates that, for this dataset, quantum models did not outperform classical approaches.

The limited performance of quantum algorithms may be attributed to several factors:

- **Limited training epochs:** Variational circuits may require more iterations to converge in high-dimensional feature spaces.
- **Low-dimensional PCA features:** Reducing to 4 principal components might have restricted the expressivity of quantum feature maps.
- **Optimization challenges:** Quantum circuits can face barren plateaus, parameter initialization sensitivity, and hardware or simulator noise.
- **Suboptimal hyperparameters:** Number of qubits, entanglement layers, and feature map repetitions might not have been optimal for the classification task.

Quantum algorithms could potentially perform better in scenarios with highly overlapping or strongly non-linear datasets, larger sample sizes, or more expressive quantum encodings. Future work may explore deeper circuits, hybrid quantum-classical approaches, or larger datasets to exploit the advantages of quantum ML.

Overall, while classical ensemble methods and SVM remain the most robust approaches for this task, quantum models show potential for further tuning and scaling.

4.4 Comparison Study

Table 3: Comparison with Previous ASD Prediction Models (Classical + Quantum)

Study	Algorithm	Acc.	Prec.	Rec.	F1
A Comparison of Machine Learning Algorithms for the Surveillance of Autism Spectrum Disorder [25]	Naive Bayes	73	–	–	78
	CVM	87	–	–	83.6
	Random Forest	87.1	–	–	86.8
An Evaluation of Machine Learning Approaches for Early Diagnosis of Autism Spectrum Disorder [26]	Random Forest	94	90.1	88.3	–
	Naive Bayes	85	84	81	–
	SVM	80	81	80	–
Towards Equitable ASD Diagnostics: A Comparative Study of Machine and Deep Learning Models Using Behavioral and Facial Data [27]	SVM	89	–	–	–
	KNN	80	–	–	–
	Gaussian NB	81	–	–	–
Machine Learning-Based Models for Early Stage Detection of Autism Spectrum Disorders [28]	Naive Bayes	96	–	–	97
	SVM	85	–	–	–
	KNN	90	–	–	–
	Random Forest	**99**	–	–	–
Detection of Autism Spectrum Disorder (ASD) in Children and Adults Using Machine Learning [29]	Logistic Regression	97.15	–	–	98
	Naive Bayes	94	–	–	96
	KNN	90.52	–	–	93
	Random Forest	81.52	–	–	88
Quantum-Based Machine Learning Approaches — Saranya & Menaka (2025) [30]	QSVM (with enhanced feature set)	98.9	–	–	–

Our approaches have been compared with some notable literary works done in the contemporary space on similar topics, and a short-hand comparison is presented in Table 3. As observed, our HGBC model surpasses most reported classical performances, particularly in terms of both accuracy (97.86%) and F1-score (97.90%). Some quantum-based approaches (e.g., QSVM) reported high accuracy up to 98.9%, but often relied on specialized features or smaller datasets.

Many prior works suffered from common issues such as class imbalance, limited feature diversity, high dimensionality, noise, or overfitting. Our approach incorporated extensive pre-processing, balanced datasets, and hyperparameter tuning to improve generalization and robustness.

The QSVM reported in [30] achieved higher accuracy (98.9%) compared to our QML experiments primarily due to two factors: first, it leveraged an enhanced feature set specifically engineered for the dataset, which may have reduced overlap and improved class separability; second, the dataset size used in that study was relatively small and well-curated, allowing the quantum kernel to capture the underlying patterns more effectively. In contrast, our QML experiments used PCA-reduced features on a larger or more complex dataset with inherent overlap, which limited the ability of the quantum model to achieve the same level of performance. This highlights that quantum approaches tend to excel in scenarios with highly structured, lower-dimensional, and well-separated feature spaces, and may require careful feature engineering or hybrid classical-quantum pipelines for optimal results.

4.5 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the distribution of ASD traits across different demographic factors within the dataset.

4.5.1 Distribution Based on Ethnicity

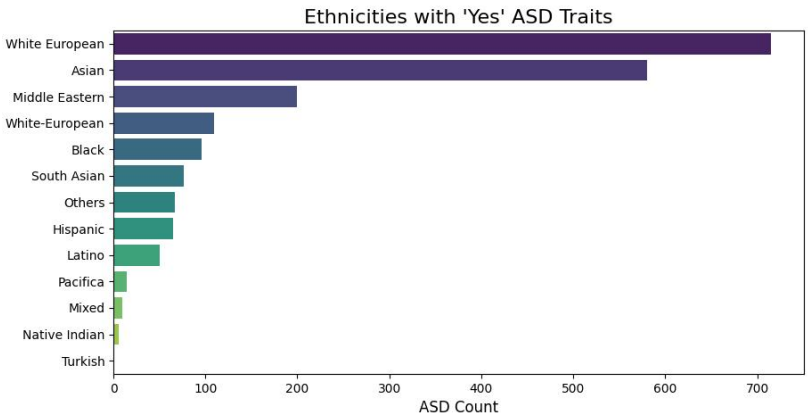

Fig. 8: The ethnics with the ASD traits as "YES"

ASD traits were found to be most prevalent in White Europeans within this dataset (Fig 8). This observation could be influenced by various factors, including genetic, social, environmental influences, and potentially higher awareness and access to diagnostic testing in these populations.

4.5.2 Distribution Based on Sex:

Analysis indicated that males exhibited ASD traits more frequently than females in this dataset (Fig 9). Hormonal differences, such as the influence of testosterone on brain development, are often considered as potential contributing factors to this observed gender disparity in ASD prevalence.

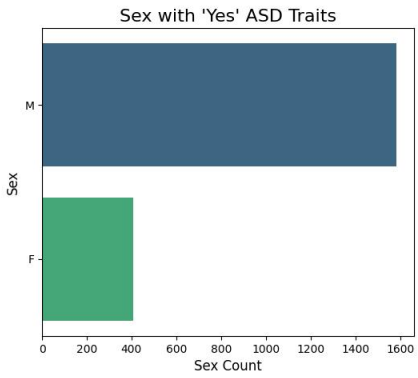

Fig. 9: Gender distribution with ASD traits as "YES"

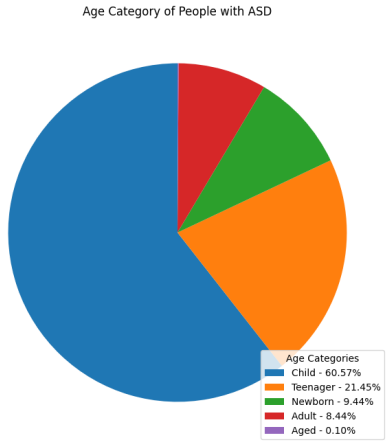

Fig. 10: Age categories distribution of ASD traits:

4.5.3 Distribution Based on Age Category

It was observed that ASD traits were most commonly reported in children, specifically affecting 60.57% of individuals in the child age category (2-12 years) within the dataset (Fig 10). This highlights a period of significant susceptibility to exhibiting these traits.

4.5.4 Distribution Based on Genetics

Analysis of family history within the dataset suggested that, for most individuals with ASD traits, other family members did not have diagnosed ASD traits (Fig 11). While genetics are known to play a role in ASD, this dataset suggests that non-genetic factors, such as environmental influences or early social experiences, may also contribute significantly to trait development.

5 Conclusion

This study successfully demonstrated the potential of supervised machine learning in predicting ASD traits using a quantitative dataset. Various classification algorithms were evaluated, with the HGBC emerging as the most effective model. HGBC achieved the highest accuracy of 97.86%, along with superior precision (98.48%), recall (97.50%), and F1-score (97.90%), indicating its strong capability in correctly identifying individuals with ASD traits while minimizing false positives. Support Vector Machine and Random Forest models also showed commendable performance.

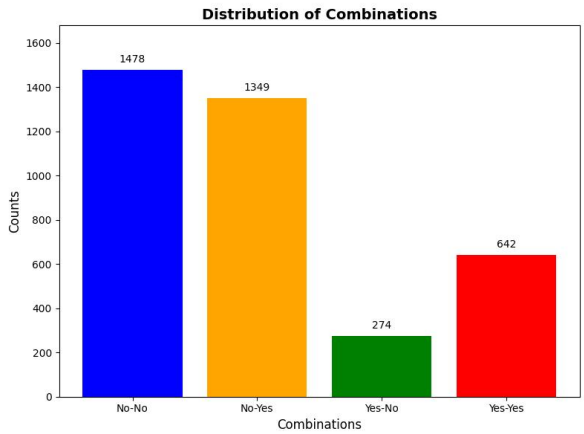

Fig. 11: The distribution of family member traits of ASD

Exploratory Data Analysis provided valuable insights into the dataset's characteristics and the distribution of ASD traits. Key findings included the high prevalence of ASD traits in children (60.57% in the child age category), the higher occurrence in males compared to females, and observations regarding ethnicity and family history, suggesting a multifactorial influence on trait manifestation.

The results highlight HGBC's potential as a valuable tool to complement traditional clinical assessments for early ASD detection. Improved early screening could facilitate timely interventions, which are known to enhance outcomes for individuals with ASD. Further advancements in machine learning techniques and the availability of more extensive datasets are expected to further improve prediction performance. The integration of these technologies into clinical workflows holds promise for developing more scalable and efficient ASD screening programs. This research represents a significant step in applying computational methods to address a critical public health challenge. Future work could involve validating the models on larger, more diverse datasets and exploring explainable AI techniques to understand the model's decision-making process better, which could further increase trust and utility in clinical settings.

References

[1] Centers for Disease Control and Prevention (2024) Autism Spectrum Disorder (ASD) - Data and Statistics. Accessed: 2025-04-20.

[2] Lord, C., Rutter, M. and Le Couteur, A. (1994) Autism Diagnostic Interview-Revised: A revised version of a diagnostic interview for caregivers of individuals with possible pervasive developmental disorders. *Journal of Autism and Developmental Disorders* **24**(5), 659–685. doi:10.1007/BF02172145

[3] Stevens, E., Dixon, D. R., Novack, M. N., Granpeesheh, D., Smith, T. and Linstead, E. (2019) Identification and analysis of behavioral phenotypes in autism spectrum disorder via unsupervised machine learning. *International Journal of Medical Informatics* **129**, 29–36. doi:10.1016/j.ijmedinf.2019.05.006

[4] Prasad, V., Sriramakrishnan, G.V. and Diana Jeba Jingle, I. (2023) Autism spectrum disorder detection using brain MRI image enabled deep learning with hybrid sewing training optimization. *Signal, Image and Video Processing* **17**, 4001–4008. doi:10.1007/s11760-023-02630-y

[5] Parlett-Pelleriti, C. M., Stevens, E., Dixon, D. et al. (2023) Applications of Unsupervised Machine Learning in Autism Spectrum Disorder Research: a Review. *Review Journal of Autism and Developmental Disorders* **10**, 406–421. doi:10.1007/s40489-021-00299-y

[6] Nogay, H. S. and Adeli, H. (2020) Machine learning (ML) for the diagnosis of autism spectrum disorder (ASD) using brain imaging. *Reviews in the Neurosciences* **31**(8), 825–841. doi:10.1515/revneuro-2020-0043

[7] Parlett-Pelleriti, C. M., Stevens, E., Dixon, D. et al. (2023) Applications of Unsupervised Machine Learning in Autism Spectrum Disorder Research: a Review. *Review Journal of Autism and Developmental Disorders* **10**, 406–421. doi:10.1007/s40489-021-00299-y

[8] Vakadkar, K., Purkayastha, D. and Krishnan, D. (2021) Detection of Autism Spectrum Disorder in Children Using Machine Learning Techniques. *SN Computer Science* **2**, 386. doi:10.1007/s42979-021-00776-5

[9] Alves, C. L., Toutain, T. G. L. d. O., de Carvalho Aguiar, P. et al. (2023) Diagnosis of Autism Spectrum Disorder Based on Functional Brain Networks and Machine Learning. *Scientific Reports* **13**, 8072. doi:10.1038/s41598-023-34650-6

[10] Mellema, C. J., Nguyen, K. P., Treacher, A. et al. (2022) Reproducible Neuroimaging Features for Diagnosis of Autism Spectrum Disorder with Machine Learning. *Scientific Reports* **12**, 3057. doi:10.1038/s41598-022-06459-2

[11] Zhou, Y., Jia, G., Ren, Y. et al. (2024) Advancing ASD Identification with Neuroimaging: A Novel GARL Methodology Integrating Deep Q-Learning and Generative Adversarial Networks. *BMC Medical Imaging* **24**, 186. doi:10.1186/s12880-024-01360-y

[12] Rajagopalan, S. S., Zhang, Y. and Yahia, A. (2024) [Title not provided – please update with article title]. *JAMA Network Open* **7**(8), e2429229. doi:10.1001/jamanetworkopen.2024.29229

[13] Jaradat, A. S., Wedyan, M., Alomari, S. and Barhoush, M. M. (2025) Using Machine Learning to Diagnose Autism Based on Eye Tracking Technology.

Diagnostics **15**(1), 66. doi:10.3390/diagnostics15010066

[14] Eslami, T. and Saeed, F. (2021) Explainable and scalable machine learning algorithms for detection of autism spectrum disorder using fMRI data. In El-Baz, A. S. and Suri, J. S. (eds.) *Neural Engineering Techniques for Autism Spectrum Disorder*, pages 39-54. Academic Press. doi:10.1016/B978-0-12-822822-7.00004-1

[15] Kotsiantis, S., Kanellopoulos, D. and Pintelas, P. (2006) Data Preprocessing for Supervised Learning. *International Journal of Computer Science* **1**, 111–117.

[16] Hosmer, D. W. and Lemeshow, S. (2000) *Applied Logistic Regression*. 2nd edition. Wiley, New York.

[17] Mienye, I. D. and Jere, N. (2024) A Survey of Decision Trees: Concepts, Algorithms, and Applications. *IEEE Access* **12**, 86716–86727. doi:10.1109/ACCESS.2024.3416838

[18] Taunk, K., De, S., Verma, S. and Swetapadma, A. (2019) A Brief Review of Nearest Neighbor Algorithm for Learning and Classification. In *2019 International Conference on Intelligent Computing and Control Systems (ICCS)*, pages 1255-1260. doi:10.1109/ICCS45141.2019.9065747

[19] Chen, H., Hu, S., Hua, R. et al. (2021) Improved naive Bayes classification algorithm for traffic risk management. *EURASIP J. Adv. Signal Process* **2021**(30).

[20] Breiman, L. (2001) Random Forests. *Machine Learning* **45**(1), 5–32. doi:10.1023/A:1010933404324

[21] Hearst, M. A., Dumais, S. T., Osuna, E., Platt, J. and Scholkopf, B. (1998) Support vector machines. *IEEE Intelligent Systems and their Applications* **13**(4), 18–28. doi:10.1109/5254.708428

[22] Maftoun, M., Shadkam, N., Komamardakhi, S. S. S., Mansor, Z. and Joloudari, J. H. (2024) Malicious URL Detection using optimized Hist Gradient Boosting Classifier based on grid search method. arXiv preprint arXiv:2404.13050.

[23] Ilemobayo, J., Durodola, O., Alade, O., Awotunde, O., Adewumi, T., Falana, O., Ogungbire, A., Osinuga, A., Ogunbiyi, D., Odezuligbo, I., Edu, O. and Ifeanyi, A. (2024) Hyperparameter Tuning in Machine Learning: A Comprehensive Review. *Journal of Engineering Research and Reports* **26**, 388–395. doi:10.9734/jerr/2024/v26i61188

[24] Bishop, C. M. (2006) *Pattern Recognition and Machine Learning*. Springer, New York, NY.

[25] Lee, Scott H. and Maenner, Matthew J. and Heilig, Charles M. *A Comparison of Machine Learning Algorithms for the Surveillance of Autism Spectrum Disorder* **Proceedings of IJ-CSE (Atlantis Press)** 10.2991/aer.k.201124.028

[26] Rasul, Rownak Ara and Saha, Promy and Bala, Diponkor and Karim, S.M. Rakib Ul and Abdullah, Md. Ibrahim and Saha, Bishwajit *An Evaluation of Machine Learning Approaches for Early Diagnosis of Autism Spectrum Disorder*

[27] Aledhari, Mohammed and Rahouti, Mohamed and Alfatemi, Ali *Towards Equitable ASD Diagnostics: A Comparative Study of Machine and Deep Learning Models Using Behavioral and Facial Data*

[28] Akter, Tania and Shahriare Satu, Md. and Khan, Md. Imran and Ali, Mohammad Hanif and Uddin, Shahadat and Lió, Pietro and Quinn, Julian M. W. and Moni, Mohammad Ali *Machine Learning-Based Models for Early Stage Detection of Autism Spectrum Disorders* **IEEE Access** 2019 166509-166527

[29] Muhammad Shoaib Farooq, Rabia Tehseen, Maidah Sabir & Zabihullah Atal *Detection of Autism Spectrum Disorder (ASD) in Children and Adults Using Machine Learning* **Scientific Reports (Nature)** 2023

[30] Saranya, S. and Menaka, R. (2025) A Quantum-Based Machine Learning Approach for Autism Detection using Common Spatial Patterns of EEG Signals. *IEEE Access* **13**, 15739–15750. doi:10.1109/ACCESS.2025.3531979